



Technology & Society: Comprehensive Midterm Notes

At a Glance:

These notes synthesize key academic concepts from your lecture materials, focusing on the foundational relationships between society and technology, social research methodologies, the evolution of culture and civilization, and the systems of social stratification.



Week 1: Society, Science, and Technology

Definition and Uses

- **Society:** An organized group of individuals within a defined territory sharing common cultural patterns, values, and institutions. It provides the social environment for nurturing human potential.
- **Science:** A systematic body of knowledge obtained through observation and experimentation to explain natural and social phenomena. It primarily answers the question "*why*".
- **Technology:** The practical application of scientific knowledge to design tools and systems that solve problems. It focuses on "*how*" to satisfy human needs.
- **Distinction:** Science is theoretical and concerned with discovering truth; technology is practical and concerned with creating useful products.

Technological Determinism

- **Driving Force:** Technology acts as a powerful agent of social change, reshaping social structures, economic activities, and cultural practices.
- **Circular Relationship:** Society generates needs that stimulate scientific research, which is then transformed into technology. These technological changes, in turn, alter social life and create new challenges.
- **Social Transformation:** In modern eras, science and technology influence how people learn, work, govern, and perceive reality.



Week 2: Social Research Methods

Basic Concepts & Methods

- **Quantitative Research:** Uses numbers and statistics, often through surveys or questionnaires, to collect data from large samples.
- **Qualitative Research:** Focuses on words and meanings through methods like in-depth interviews or participant observation (ethnography) to provide rich, detailed descriptions.
- **Mixed Methods:** Many studies combine both approaches for a more comprehensive

understanding of social issues.

Stages of Social Research (6–8 Step Lifecycle)

- **Topic Selection:** Choosing a relevant issue or defining a problem.
- **Review of Literature:** Reading existing studies to build on current knowledge.
- **Formulating Questions/Hypotheses:** Turning topics into clear, testable questions.
- **Research Design:** Planning methods, sample size, and ethical considerations.
- **Sampling:** Selecting a representative subset of the population.
- **Data Collection:** Gathering data systematically through fieldwork.
- **Data Analysis:** Organizing data (e.g., using SPSS for quantitative) and finding patterns.
- **Reporting:** Drawing conclusions and sharing findings for policy or academic use.

Week 3: Culture and Civilization

Culture vs. Civilization

- **Culture:** The "software" of society, including shared patterns of behavior, beliefs, and values. It consists of *material culture* (tangible objects like tools and technology) and *non-material culture* (intangible ideas, laws, and morals).
- **Civilization:** A more advanced stage of cultural development marked by urbanization, writing systems, and centralized governance. It is the "grand edifice" built on the foundation of culture.

Cultural Lag

- **Definition:** A phenomenon where material culture (technology) advances faster than non-material culture (laws, morals, or norms).
- **AI/Technology:** We build advanced tools but struggle to establish ethical rules and regulations.
- **Social Media:** Instant global connectivity exists, but legal frameworks for privacy and misinformation are still catching up.
- **Consequences:** Can lead to social tension, conflict, or maladjustment until non-material aspects adapt.

Week 4: Socialization and Personality

Socialization

- **Primary Socialization:** Early, intimate learning during childhood, primarily through the family, which teaches basic skills and core values.
- **Secondary Socialization:** Later, more formal learning through schools, peer groups, mass media, and religion.

Personality Development

- **Nature vs. Nurture:** A fundamental debate in sociology. Nature refers to

biological/genetic predispositions, while Nurture refers to environmental influences and socialization. Your materials emphasize nurture—how agents of socialization shape how individuals think and act.

- **Peer Groups:** Provide an arena for identity exploration and learning group norms.
- **Mass Media:** Shapes perceptions of reality, beauty standards, and success.



Week 5: Stratification and Inequality

Social Stratification

- **Definition:** The patterned arrangement of roles and statuses in society, creating distinct layers based on three main dimensions (often associated with Max Weber):
 - **Class (Economic):** Wealth and income.
 - **Status (Social):** Prestige, honor, and respect.
 - **Power (Political):** Ability to enforce one's will over others.
- **Dimensions of Status:** Status can be *ascribed* (assigned at birth, e.g., race, royal lineage) or *achieved* (earned through effort and merit, e.g., education, career).

Systems of Stratification

- **Closed Systems:** Low mobility where status is ascribed (e.g., Slavery, Caste, or Estate systems).
- **Open Systems:** Higher mobility based primarily on achieved status (e.g., Class systems in modern industrial societies).

Social Inequality & Technology

- **Digital Divide:** Technology does not reach everyone equally; urban and educated populations often benefit more, while remote or poor communities struggle with access. This gap can worsen existing social inequalities.

💡 *Exam Pro Tip: In Technology & Society, examiners love real-world examples. When defining concepts like Cultural Lag or the Digital Divide, always cite modern examples (e.g., AI regulation, CRISPR, or remote learning during the pandemic) to secure full marks!*

